

ARITMÉTICA - ÁLGEBRA PREFACULTATIVO 2-2025
SOLUCIONARIO PRIMER PARCIAL

1. $a + b + c = 5$ y

$$\begin{aligned} E &= a(a + 2b - 3) + b(b + 2c - 3) + c(c + 2a - 3) \\ E &= a^2 + 2ab + 2ac - 3a + b^2 + 2bc - 3b + c^2 - 3c \\ &= a^2 + b^2 + c^2 + 2ab + 2ac + 2bc - 3a - 3b - 3c \\ &= (a + b + c)^2 - 3(a + b + c) = 5^2 - 3(5) = 10 \rightarrow E = 10 \rightarrow \text{resp. (b)} \end{aligned}$$

$$2. \begin{cases} 12(x + y) = 5xy \\ 18(y + z) = 5yz \\ 36(x + z) = 13xz \end{cases} \rightarrow \begin{cases} \frac{(x + y)}{xy} = \frac{5}{12} \\ \frac{(y + z)}{yz} = \frac{5}{18} \\ \frac{(x + z)}{xz} = \frac{13}{36} \end{cases} \rightarrow \begin{cases} \frac{1}{y} + \frac{1}{x} = \frac{5}{12} \quad (1) \\ \frac{1}{z} + \frac{1}{y} = \frac{5}{18} \quad (2) \\ \frac{1}{z} + \frac{1}{x} = \frac{13}{36} \quad (3) \end{cases}$$

$$(1)-(2): \frac{1}{x} - \frac{1}{z} = \frac{5}{12} - \frac{5}{18} \rightarrow \frac{1}{x} - \frac{1}{z} = \frac{5}{36} \quad (4)$$

$$(3)+(4): \frac{2}{x} = \frac{18}{36} \rightarrow \boxed{x = 4} \rightarrow \text{en (1): } \frac{1}{y} + \frac{1}{4} = \frac{5}{12} \rightarrow \boxed{y = 6}$$

$$\text{En (2): } \frac{1}{z} + \frac{1}{6} = \frac{5}{18} \rightarrow \boxed{z = 9} \rightarrow x + y + z = 4 + 6 + 9 = 19 \rightarrow \text{resp. (d)}$$

$$\begin{aligned} 3. & \frac{4x-5}{x^2+x-12} + \frac{9}{18-3x-x^2} + \frac{2}{x^2+10x+24} \\ &= \frac{4x-5}{(x+4)(x-3)} - \frac{9}{(x+6)(x-3)} + \frac{2}{(x+6)(x+4)} = \frac{(4x-5)(x+6) - 9(x+4) + 2(x-3)}{(x+4)(x-3)(x+6)} \\ &= \frac{4x^2 + 12x - 72}{(x+4)(x-3)(x+6)} = \frac{4(x+6)(x-3)}{(x+4)(x-3)(x+6)} = \frac{4}{x+4} \rightarrow \text{resp. (b)} \end{aligned}$$

4. El número de divisores de $2^a 7^b$ es $(a+1)(b+1)$

El número de divisores de $\frac{2^a 7^b}{4} = 2^{a-2} 7^b$ es $(a-2+1)(b+1)$

El número de divisores de $14(2^a 7^b) = 2^{a+1} 7^{b+1}$ es $(a+1+1)(b+1+1)$

$$(a-2+1)(b+1) = \frac{(a+1)(b+1)}{3} \rightarrow 3(a-1) = a+1 \rightarrow a = 2$$

$$(a+1+1)(b+1+1) = 2(a+1)(b+1) \rightarrow \text{Reemplazamos } a = 2$$

$$(4)(b+2) = 2(3)(b+1) \rightarrow b = 1 \rightarrow 2^2 7^1 = 28 \rightarrow \text{resp. (c)}$$

	OBREROS	DÍAS	HORAS/DÍA	LITROS	
5.	15	40	10	80000	$\rightarrow \frac{40}{X} = \left(\frac{10}{15}\right)\left(\frac{6}{10}\right)\left(\frac{80000}{40000}\right),$
	10	X	6	40000	
	INVERSA		INVERSA	DIRECTA	

$$\rightarrow X = 50 \text{ DÍAS} \rightarrow \text{resp. (a)}$$